

Algebra

CHAPTER 11

Statistics

11.1–11.10

Measures of Central Tendency	11.1
Mean	11.1
Median	11.3
Mode	11.4
Relation between Mean, Mode and Median	11.4
<i>Concept Application Exercise 11.1</i>	<i>11.4</i>
Measures of Dispersion	11.5
Range	11.5
Mean Deviation	11.5
Variance	11.5
Standard Deviation (S.D.)	11.6
<i>Concept Application Exercise 11.2</i>	<i>11.7</i>
<i>Exercises</i>	<i>11.8</i>
<i>Single Correct Answer Type</i>	<i>11.8</i>
<i>Answers Key</i>	<i>11.10</i>

MEASURES OF CENTRAL TENDENCY

MEAN

Arithmetic Mean

The mean is the sum of the value of each observation in a dataset divided by the number of observations.

Mean of ungrouped data

If the n observations in data are $x_1, x_2, x_3, \dots, x_n$; then the arithmetic mean $\bar{x}$ is given by

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$

Shortcut method

$$\text{Arithmetic mean } \bar{x} = A + \frac{\sum_{i=1}^n d_i}{n},$$

where, A = assumed mean, d_i = deviation from assumed mean
 $= x_i - A$

Mean of discrete frequency distribution and continuous frequency distribution

If $x_1, x_2, \dots, x_n$ are observations with respective frequencies $f_1, f_2, \dots, f_n$; then

Sum of the values of the observation = $f_1x_1 + f_2x_2 + \dots + f_nx_n$

and Number of observations = $f_1 + f_2 + \dots + f_n$

The mean in this case is given by

$$\bar{x} = \frac{f_1x_1 + f_2x_2 + f_3x_3 + \dots + f_nx_n}{f_1 + f_2 + f_3 + \dots + f_n} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i}$$

In continuous frequency distribution, x_i is taken as mid-point of respective classes.

Shortcut method

Arithmetic mean, $\bar{x} = A + \frac{\sum_{i=1}^n f_i (x_i - A)}{n}$, where A is assumed mean.

Combined mean

If set of N_1 observations has mean $\bar{X}_1$ and set of N_2 observations has mean $\bar{X}_2$, then their combined mean is $\frac{N_1\bar{X}_1 + N_2\bar{X}_2}{N_1 + N_2}$.

Note:

- The sum of the squares of the deviations from the mean is minimum, i.e., $\sum_{i=1}^n (x_i - \bar{x})^2$ is least.
- The sum of the deviations of items from their mean is equal to zero, i.e., $\sum_{i=1}^n (x_i - \bar{x}) = 0$.
- The mean is affected accordingly if the observations are given a mathematical treatment by any constant item.
- The arithmetic mean is independent of the origin, i.e., it is not affected by any change in origin (assumed mean).

Geometric Mean

If the n observations in the data are $x_1, x_2, x_3, \dots, x_n$; then the geometric mean is given by

$$\text{G.M.} = (x_1 \cdot x_2 \cdot x_3 \cdot \dots \cdot x_n)^{1/n}$$

If $x_1, x_2, \dots, x_n$ are observations with respective frequencies $f_1, f_2, \dots, f_n$; then geometric mean is given by

$$\text{G.M.} = (x_1^{f_1} \cdot x_2^{f_2} \cdot \dots \cdot x_n^{f_n})^{1/N}, \text{ where } N = f_1 + f_2 + \dots + f_n.$$

Harmonic Mean

The harmonic mean of n observations $x_1, x_2, x_3, \dots, x_n$ is

$$\text{H.M.} = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}}$$

If $x_1, x_2, \dots, x_n$ are observations with respective frequencies $f_1, f_2, \dots, f_n$, then harmonic mean is given by

$$\text{H.M.} = \frac{f_1 + f_2 + f_3 + \dots + f_n}{\frac{f_1}{x_1} + \frac{f_2}{x_2} + \dots + \frac{f_n}{x_n}}$$

ILLUSTRATION 11.1

If a variate X is expressed as a linear function of two variates U and V in the form $X = aU + bV$, then the mean $\bar{X}$ of X is

- (1) $a\bar{U} - b\bar{V}$ (2) $\bar{U} + \bar{V}$
 (3) $b\bar{U} + a\bar{V}$ (4) None of these

Sol.

(4) We have $\Sigma X = a\Sigma U + b\Sigma V$. Therefore,

$$\bar{X} = \frac{1}{n} \Sigma X = a \cdot \frac{1}{n} \Sigma U + b \cdot \frac{1}{n} \Sigma V = a\bar{U} + b\bar{V}$$

ILLUSTRATION 11.2

If the mean of the numbers $27 + x$, $31 + x$, $89 + x$, $107 + x$, $156 + x$ is 82, then the mean of $130 + x$, $126 + x$, $68 + x$, $50 + x$, $1 + x$ is

- (1) 75 (2) 157 (3) 82 (4) 80

Sol.

(1) Given

$$82 = \frac{(27+x) + (31+x) + (89+x) + (107+x) + (156+x)}{5}$$

$$\Rightarrow 82 \times 5 = 410 + 5x$$

$$\Rightarrow 410 - 410 = 5x$$

$$\Rightarrow x = 0$$

Therefore, the required mean is

$$\begin{aligned} \bar{x} &= \frac{130 + x + 126 + x + 68 + x + 50 + x + 1 + x}{5} \\ &= \frac{375 + 5x}{5} = \frac{375 + 0}{5} = \frac{375}{5} = 75 \end{aligned}$$

ILLUSTRATION 11.3

If the arithmetic mean of the numbers $x_1, x_2, x_3, \dots, x_n$ is $\bar{x}$, then the arithmetic mean of the numbers $ax_1 + b, ax_2 + b, ax_3 + b, \dots, ax_n + b$, where a, b are two constants, would be

- (1) $\bar{x}$ (2) $na\bar{x} + nb$ (3) $a\bar{x}$ (4) $a\bar{x} + b$

Sol.

(4) Required mean

$$\begin{aligned} &= \frac{(ax_1 + b) + (ax_2 + b) + \dots + (ax_n + b)}{n} \\ &= \frac{a(x_1 + x_2 + \dots + x_n)}{n} + b = a\bar{x} + b \end{aligned}$$

($\because \frac{x_1 + x_2 + \dots + x_n}{n} = \bar{x}$)

ILLUSTRATION 11.4

The weighted mean of the first n natural numbers whose weights are equal to the squares of the corresponding numbers is

- (1) $\frac{n+1}{2}$ (2) $\frac{3n(n+1)}{2(2n+1)}$
 (3) $\frac{(n+1)(2n+1)}{6}$ (4) $\frac{n(n+1)}{2}$

Sol.

$$\begin{aligned} (2) \text{ Weighted mean} &= \frac{1 \cdot 1^2 + 2 \cdot 2^2 + \dots + n \cdot n^2}{1^2 + 2^2 + \dots + n^2} \\ &= \frac{\Sigma n^3}{\Sigma n^2} = \frac{\frac{n(n+1)}{2} \cdot \frac{n(n+1)}{2}}{\frac{n(n+1)(2n+1)}{6}} = \frac{3n(n+1)}{2(2n+1)} \end{aligned}$$

ILLUSTRATION 11.5

A student obtain 75%, 80% and 85% in three subjects. If the marks of another subject is added, then his average cannot be less than

- (1) 60% (2) 65% (3) 80% (4) 90%

Sol.

(1) Marks obtained from three subjects out of 300 is

$$75 + 80 + 85 = 240$$

If the marks of another subject is added, then the marks will be ≥ 240 out of 400.

$$\therefore \text{Minimum average marks} = \frac{240}{4} = 60\%$$

[When marks in the fourth subject = 0]

ILLUSTRATION 11.6

If $\bar{x}_1$ and $\bar{x}_2$ are the means of two distributions such that $\bar{x}_1 < \bar{x}_2$ and $\bar{x}$ is the mean of the combined distribution, then

- (1) $\bar{x} < \bar{x}_1$ (2) $\bar{x} > \bar{x}_2$
 (3) $\bar{x} = \frac{\bar{x}_1 + \bar{x}_2}{2}$ (4) $\bar{x}_1 < \bar{x} < \bar{x}_2$

Sol.

(4) Let n_1 and n_2 be the number of observations in two groups having means $\bar{x}_1$ and $\bar{x}_2$, respectively. Then,

$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$$

$$\begin{aligned} \text{Now, } \bar{x} - \bar{x}_1 &= \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2} - \bar{x}_1 \\ &= \frac{n_2(\bar{x}_2 - \bar{x}_1)}{n_1 + n_2} > 0 \quad [\because \bar{x}_2 > \bar{x}_1] \end{aligned}$$

$$\Rightarrow \bar{x} > \bar{x}_1, \quad (1)$$

$$\text{and } \bar{x} - \bar{x}_2 = \frac{n_1(\bar{x}_1 - \bar{x}_2)}{n_1 + n_2} < 0 \quad [\because \bar{x}_2 > \bar{x}_1]$$

$$\Rightarrow \bar{x} < \bar{x}_2 \quad (2)$$

From (1) and (2), $\bar{x}_1 < \bar{x} < \bar{x}_2$.

ILLUSTRATION 11.7

The mean of 100 items is 49. It was discovered that three items which should have been 60, 70, 80 were wrongly read as 40, 20, 50, respectively. The correct mean is

- (1) 48 (2) 82.5 (3) 50 (4) 80

Sol.

(3) Sum of 100 items = $49 \times 100 = 4900$

$$\text{Sum of items added} = 60 + 70 + 80 = 210$$

$$\text{New sum} = 4900 + 210 - 110 = 5000$$

$$\therefore \text{Correct mean} = \frac{5000}{100} = 50$$

ILLUSTRATION 11.8

The mean weight per student in a group of seven students is 55 kg. If the individual weights of six students are 52, 58, 55, 53, 56 and 54, then the weight of the seventh student is

- (1) 55 kg (2) 60 kg (3) 57 kg (4) 50 kg

Sol.

- (3) The total weight of seven students is $55 \times 7 = 385$ kg

The sum of the weights of six students is

$$52 + 58 + 55 + 53 + 56 + 54 = 328 \text{ kg}$$

Hence, the weight of the seventh student is $385 - 328 = 57$ kg

MEDIAN

The median is the middle value (or central value) in distribution when the values are arranged in ascending or descending order.

Median divides the distribution in two equal halves (i.e., there are 50% of observations on either side of the median value).

Median of Ungrouped Data

Let n be the number of observations.

To find the median, first arrange the observations in ascending or descending order.

If n is odd,

$$\text{Median} = \text{Value of } \left(\frac{n+1}{2}\right)^{\text{th}} \text{ observation}$$

If n is even,

$$\begin{aligned} &\text{Value of } \left(\frac{n}{2}\right)^{\text{th}} \text{ observation} \\ &+ \text{Value of } \left(\frac{n}{2} + 1\right)^{\text{th}} \text{ observation} \\ \text{Median} = &\frac{\quad}{2} \end{aligned}$$

- Median of discrete frequency distribution**

To find the median, the observations are arranged in ascending order. After this the cumulative frequencies are obtained. Then, we identify the observation whose cumulative frequency is equal to or just greater than $\frac{N}{2}$,

where N is the sum of frequencies. This value of the observation lies in the middle of the data and therefore, it is the required median.

- Median of continuous frequency distribution**

In this case, following formula can be used:

(a) For series in ascending order,

$$\text{Median} = l + \frac{\left(\frac{N}{2} - C\right)}{f} \times i;$$

where, l = Lower limit of the median class
 f = Frequency of the median class
 N = The sum of all frequencies

i = The width of the median class

C = The cumulative frequency of the class preceding to median class.

(b) For series in descending order,

$$\text{Median} = u - \left(\frac{\frac{N}{2} - C}{f}\right) \times i$$

where u = upper limit of the median class, $N = \sum_{i=1}^n f_i$.

Note:

- Median is “positional” average. It is a measure of position rather than of magnitude.
- It is a positional average and it is not influenced by the position of the items.
- Extreme values do not affect the median as strongly as they affect the mean.
- The sum of the absolute values of the deviations of the items from the median is minimum.

ILLUSTRATION 11.9

If a variable takes the discrete values $\alpha + 4$,

$$\alpha - \frac{7}{2}, \alpha - \frac{5}{2}, \alpha - 3, \alpha - 2, \alpha + \frac{1}{2}, \alpha - \frac{1}{2}, \alpha + 5 (\alpha > 0),$$

then the median is

- (1) $\alpha - \frac{5}{4}$ (2) $\alpha - \frac{1}{2}$ (3) $\alpha - 2$ (4) $\alpha + \frac{5}{4}$

Sol.

- (1) Arrange the data as follows:

$$\alpha - \frac{7}{2}, \alpha - 3, \alpha - \frac{5}{2}, \alpha - 2, \alpha - \frac{1}{2}, \alpha + \frac{1}{2}, \alpha + 4, \alpha + 5$$

$$\text{Median} = \frac{1}{2} [\text{value of 4th item} + \text{value of 5th item}]$$

$$\therefore \text{Median} = \frac{\alpha - 2 + \alpha - \frac{1}{2}}{2} = \frac{2\alpha - \frac{5}{2}}{2} = \alpha - \frac{5}{4}$$

ILLUSTRATION 11.10

The median of a set of nine distinct observations is 20.5. If each of the last four observations of the set is increased by 2, then the median of the new set

- (1) Is increased by 2
 (2) Is decreased by 2
 (3) Is two times the original median
 (4) Remains the same as that of the original set

Sol.

- (4) Since $n = 9$, median term = $\left(\frac{9+1}{2}\right)^{\text{th}} = 5^{\text{th}}$ term.

Now, the last four observations are increased by 2.

Since the median is the 5th observation, which remains unchanged, there will be no change in median.

MODE

For discrete frequency distribution, mode is that value of the observation which occurs most frequently, i.e., an observation with the maximum frequency is called the mode. It is possible that more than one values may have the same maximum frequency. In such situations, the data is said to be multimodal. But here, we shall restrict ourselves to problems having a single mode only.

In a continuous frequency distribution, it is not possible to determine the mode by looking at the frequencies. Here, we can only locate a class with the maximum frequency, called the modal class. The mode is a value inside the modal class, and is given by the formula:

Mode = $l + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times i$

Where,

l = The lower limit of the modal class

f_1 = The frequency of the modal class

f_0 = The frequency of the class preceding the modal class

f_2 = The frequency of the class succeeding the modal class

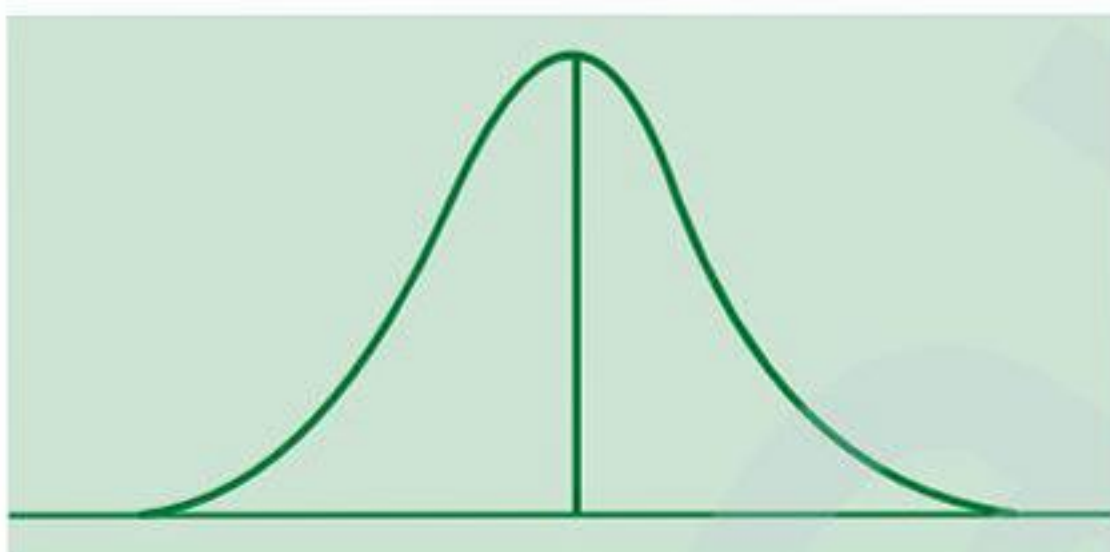
i = The size of the modal class.

- Note:**
- Mode is not affected by the presence of extremely large or small items.
 - Mode has an advantage over the median and the mean as it can be found for both numerical and categorical (non-numerical) data.

RELATION BETWEEN MEAN, MODE AND MEDIAN

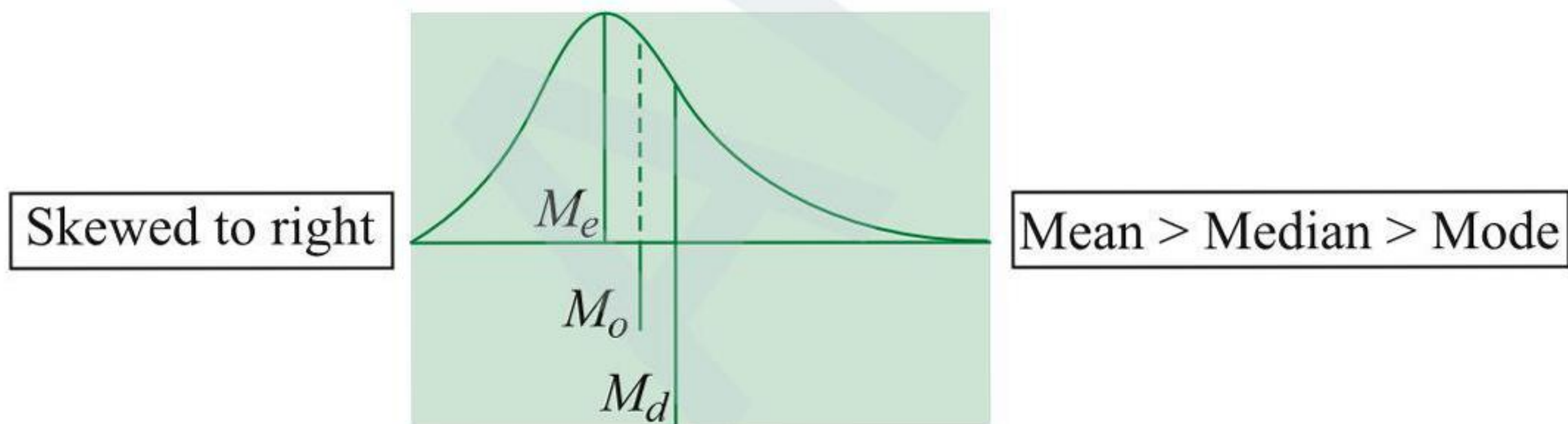
If a distribution is a symmetric distribution, the values of mean, mode and median coincide.

In a symmetric distribution, frequencies are symmetrically distributed on both sides of the centre point of the frequency curve.

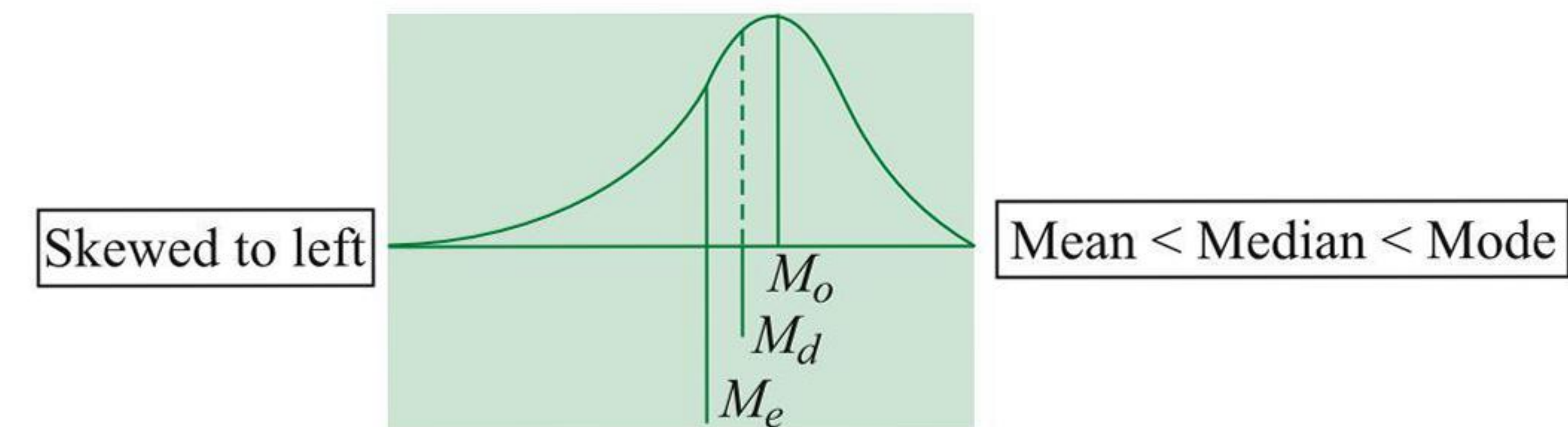


Distribution which is not symmetric is called a skewed-distribution.

If a frequency distribution is positively skewed, then mean is greater than median and median is greater than mode.



If a frequency distribution is negatively skewed, then mean is less than median and median is less than mode.



In a moderately asymmetric distribution, the interval between the mean and the median is approximately one-third of the interval between the mean and the mode.

We have the following empirical relation between them:

Mean – Mode = 3(Mean – Median)

⇒ Mode = 3 Median – 2 Mean

It is known as empirical relation.

ILLUSTRATION 1 1.1 1

If in a moderately asymmetrical distribution the mode and the mean of the data are 6λ and 9λ , respectively, then the median is

(1) 8λ (2) 7λ (3) 6λ (4) 5λ

Sol.

(1) For a moderately skewed distribution,

Mode = 3 median – 2 mean

⇒ $6\lambda = 3 \text{ median} - 18\lambda \Rightarrow \text{median} = 8\lambda$

CONCEPT APPLICATION EXERCISE 1 1.1

1. In a class of 100 students, there are 70 boys whose average marks in a subject is 75. If the average marks of the complete class is 72, then what is the average of the girls?
(1) 73 (2) 65 (3) 68 (4) 74
2. The median of a set of nine distinct observations is 20.5. If each of the largest four observations of the set is increased by 2, then the median of the new set
(1) is increased by 2.
(2) is decreased by 2.
(3) is two times the original median.
(4) remains the same as that of the original set.
3. If in a frequency distribution, the mean and median are 21 and 22, respectively, then its mode is approximately
(1) 22.0 (2) 20.5 (3) 25.5 (4) 24.0
4. The average marks of boys in a class is 52 and that of girls is 42. The average marks of boys and girls combined is 50. The percentage of boys in the class is
(1) 40 (2) 20 (3) 80 (4) 60
5. Compute the median from the following table

Marks obtained	No. of students
0–10	2
10–20	18
20–30	30
30–40	45
40–50	35
50–60	20
60–70	6
70–80	3

- (1) 36.55 (2) 35.55
(3) 40.05 (4) None of these

ANSWERS

1. (2) 2. (4) 3. (4) 4. (3) 5. (1)

MEASURES OF DISPERSION

The degree to which numerical data tends to spread about an average value is called the dispersion of the data.

The four measures of dispersion are

- (a) Range
- (b) Mean deviation
- (c) Standard deviation
- (d) Square deviation

RANGE

Range is the difference between the highest and the lowest values in a series (observations).

The range of data gives us a rough idea of variability or scatter but does not tell about the dispersion of the data from a measure of central tendency.

MEAN DEVIATION

Mean Deviation of Ungrouped Data

If the n observations in the data are $x_1, x_2, x_3, \dots, x_n$; then

Mean deviation about any number 'a', $M.D.(a) = \frac{1}{n} \sum_{i=1}^n |x_i - a|$

Mean deviation about mean, $M.D.(\bar{x}) = \frac{1}{n} \sum_{i=1}^n |x_i - \bar{x}|$

Mean deviation about median, $M.D.(M) = \frac{1}{n} \sum_{i=1}^n |x_i - M|$

Mean Deviation of Discrete Frequency Distribution

If $x_1, x_2, \dots, x_n$ are observations with respective frequencies $f_1, f_2, \dots, f_n$, then

Mean deviation about mean is

$$M.D.(\bar{x}) = \frac{1}{N} \sum_{i=1}^n f_i |x_i - \bar{x}|,$$

where $N = \sum_{i=1}^n f_i$ and $\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i}$

Mean deviation about median is

$$M.D.(M) = \frac{1}{N} \sum_{i=1}^n f_i |x_i - M|$$

Mean Deviation of Continuous Frequency Distribution

The formulas for mean deviation are same as in the case of discrete frequency distribution. Here, x_i is the midpoint of each class.

ILLUSTRATION 11.12

The mean deviation about the mean of the following distribution is

Size	20	21	22	23	24
Frequency	6	4	5	1	4

- (1) 1 (2) 1.25 (3) 1.5 (4) 1.75

Sol. (2)

Size (x_i)	Frequency (f_i)	$f_i x_i$	$d_i = x_i - \bar{x} $	$f_i d_i$
20	6	120	1.65	9.90
21	4	84	0.65	2.60
22	5	110	0.35	1.75
23	1	23	1.35	1.35
24	4	96	2.35	9.40
Total	20	433		25

$$\text{Now, } \bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{433}{20} = 21.65$$

$$\therefore M.D. = \frac{\sum f_i |x_i - \bar{x}|}{\sum f_i} = \frac{25}{20} = 1.25$$

ILLUSTRATION 11.13

The mean deviation about the median of the following distribution is

Marks obtained	10	11	12	14	15
Number of students	2	3	8	3	4

- (1) 1 (2) 1.25 (3) 1.5 (4) 1.75

Sol. (2)

Marks obtained (x_i)	f_i	c.f.	$d_i = x_i - M_e $	$f_i d_i$
10	2	2	2	4
11	3	5	1	3
12	8	13	0	0
14	3	16	2	6
15	4	20	3	12
Total	$\sum f_i = 20$			$\sum f_i d_i = 25$

$$\text{Now, } M_e = \left(\frac{20+1}{2} \right) \text{th item} = \left(\frac{21}{2} \right) = 10.5 \text{th item}$$

$$\therefore M_e = 12$$

$$\therefore M.D. = \frac{\sum f_i d_i}{\sum f_i} = \frac{25}{20} = 1.25$$

VARIANCE

The arithmetic mean of the squares of the deviations of the variable values from their actual arithmetic mean is known as variance (σ^2).

For ungrouped data,

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - \left(\frac{1}{n} \sum_{i=1}^n x_i \right)^2$$

For discrete frequency distribution and continuous frequency distribution,

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^2 = \frac{1}{N} \sum_{i=1}^n f_i x_i^2 - \left(\frac{1}{N} \sum_{i=1}^n f_i x_i \right)^2,$$

where $N = \sum_{i=1}^n f_i$

Here, in the case of continuous frequency distribution, x_i is the midpoint of each class.

ILLUSTRATION 11.14

The variance of the first n natural numbers is

- (1) $\frac{n^2 - 1}{12}$ (2) $\frac{n^2 - 1}{6}$
 (3) $\frac{n^2 + 1}{6}$ (4) $\frac{n^2 + 1}{12}$

Sol.

$$\begin{aligned} \text{(1) Variance} &= (SD)^2 = \frac{1}{n} \sum x^2 - \left(\frac{\sum x}{n} \right)^2, \quad \left(\because \bar{x} = \frac{\sum x}{n} \right) \\ &= \frac{n(n+1)(2n+1)}{6n} - \left(\frac{n(n+1)}{2n} \right)^2 = \frac{n^2 - 1}{12} \end{aligned}$$

ILLUSTRATION 11.15

The mean of five observations is 4 and their variance is 5.2. If three of these observations are 1, 2 and 6, then the other two are

- (1) 2 and 9 (2) 3 and 8
 (3) 4 and 7 (4) 5 and 6

Sol.

(3) Let the two unknown items be x and y , then

$$\text{Mean} = 4 \Rightarrow \frac{1+2+6+x+y}{5} = 4$$

$$\Rightarrow x + y = 11 \quad (1)$$

and variance = 5.2

$$\Rightarrow \frac{1^2 + 2^2 + 6^2 + x^2 + y^2}{5} - (\text{Mean})^2 = 5.2$$

$$41 + x^2 + y^2 = 5[5.2 + (4)^2]$$

$$41 + x^2 + y^2 = 106$$

$$x^2 + y^2 = 65 \quad (2)$$

Solving (1) and (2) for x and y , we get

$$x = 4, y = 7 \text{ or } x = 7, y = 4$$

STANDARD DEVIATION (S.D.)

Non-negative square root of variance is called standard deviation and it is represented by σ .

For ungrouped data,

$$\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2} = \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2 - \left(\frac{1}{n} \sum_{i=1}^n x_i \right)^2}$$

For discrete frequency distribution and continuous frequency distribution,

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^2} = \sqrt{\frac{1}{N} \sum_{i=1}^n f_i x_i^2 - \left(\frac{1}{N} \sum_{i=1}^n f_i x_i \right)^2},$$

where $N = \sum_{i=1}^n f_i$

Here, in the case of continuous frequency distribution, x_i is the midpoint of each class.

ILLUSTRATION 11.16

The standard deviation of the data 6, 5, 9, 13, 12, 8, 10 is

- (1) $\sqrt{\frac{52}{7}}$ (2) $\frac{52}{7}$ (3) $\sqrt{6}$ (4) 6

Sol.

(1) Given, data are 6, 5, 9, 13, 12, 8, and 10

x_i	x_i^2
6	36
5	25
9	81
13	169
12	144
8	64
10	100
$\Sigma x_i = 63$	$\Sigma x_i^2 = 619$

$$\begin{aligned} \therefore \text{S.D.} = \sigma &= \sqrt{\frac{\Sigma x_i^2}{N} - \left(\frac{\Sigma x_i}{N} \right)^2} = \sqrt{\frac{619}{7} - \left(\frac{63}{7} \right)^2} \\ &= \sqrt{\frac{7 \times 619 - 3969}{49}} \\ &= \sqrt{\frac{4333 - 3969}{49}} \\ &= \sqrt{\frac{364}{49}} = \sqrt{\frac{52}{7}} \end{aligned}$$

ILLUSTRATION 11.17

Consider the frequency distribution, where A is a positive integer:

x	A	$2A$	$3A$	$4A$	$5A$	$6A$
f	2	1	1	1	1	1

If the variance is 160, then the value of A is

- (1) 5 (2) 6 (3) 7 (4) 8

Sol. (3)

x	f_i	$f_i x_i$	$f_i x_i^2$
A	2	$2A$	$2A^2$
$2A$	1	$2A$	$4A^2$
$3A$	1	$3A$	$9A^2$
$4A$	1	$4A$	$16A^2$
$5A$	1	$5A$	$25A^2$
$6A$	1	$6A$	$36A^2$
Total	$n = 7$	$\Sigma f_i x_i = 22A$	$\Sigma f_i x_i^2 = 92A^2$

$$\begin{aligned}
\therefore \sigma^2 &= \frac{\sum f_i x_i^2}{n} - \left(\frac{\sum f_i x_i}{n} \right)^2 \\
\Rightarrow 160 &= \frac{92A^2}{7} - \left(\frac{22A}{7} \right)^2 \\
\Rightarrow 160 &= \frac{92A^2}{7} - \frac{484A^2}{49} \\
\Rightarrow 160 &= (644 - 484) \frac{A^2}{49} \\
\Rightarrow 160 &= \frac{160A^2}{49} \\
\Rightarrow A^2 &= 49 \\
\therefore A &= 7
\end{aligned}$$

ILLUSTRATION 11.18

The standard deviation of the following frequency distribution is

X	2	3	4	5	6	7
f	4	9	16	14	11	6

- (1) 1.38 (2) 1.42 (3) 1.45 (4) 1.60
Find the standard deviation.

Sol. (1)

x_i	f_i	$d_i = x_i - 4$	$f_i d_i$	$f_i d_i^2$
2	4	-2	-8	16
3	9	-1	-9	9
4	16	0	0	0
5	14	1	14	14
6	11	2	22	44
7	6	3	18	54
Total	60		$\sum f_i d_i = 37$	$\sum f_i d_i^2 = 137$

$$\begin{aligned}
\therefore \text{S.D.} &= \sqrt{\frac{\sum f_i d_i^2}{n} - \left(\frac{\sum f_i d_i}{n} \right)^2} \\
&= \sqrt{\frac{137}{60} - \left(\frac{37}{60} \right)^2} = \sqrt{2.2833 - (0.616)^2} \\
&= \sqrt{2.2833 - 0.3794} = \sqrt{1.9037} = 1.38
\end{aligned}$$

ILLUSTRATION 11.19

Let a, b, c, d and e be the observations with mean m and standard deviation s . The standard deviation of the observations $a + k, b + k, c + k, d + k, e + k$ is

- (1) ks (2) s (3) $s + k$ (4) $\frac{s}{k}$

Sol.

- (2) Given observations are a, b, c, d and e .

$$\text{Mean} = m = \frac{a + b + c + d + e}{5}$$

$$\sum x_i = a + b + c + d + e = 5m$$

$$\begin{aligned}
\text{New mean} &= \frac{a + k + b + k + c + k + d + k + e + k}{5} \\
&= \frac{(a + b + c + d + e) + 5k}{5} = m + k
\end{aligned}$$

$$\begin{aligned}
\therefore \text{S.D.} &= \sqrt{\frac{\sum (x_i^2 + k^2 + 2kx_i)}{n} - (m^2 + k^2 + 2mk)} \\
&= \sqrt{\frac{\sum x_i^2}{n} - m^2 + \frac{2k \sum x_i}{n} - 2mk} \\
&= \sqrt{\frac{\sum x_i^2}{n} - m^2 + 2km - 2mk} \quad \left[\because \frac{\sum x_i}{n} = m \right] \\
&= \sqrt{\frac{\sum x_i^2}{n} - m^2} \\
&= s
\end{aligned}$$

CONCEPT APPLICATION EXERCISE 11.2

1. In an experiment with 15 observations on x , the following results were available:

$$\sum x^2 = 2830, \sum x = 170$$

One observation, 20, was found to be wrong and was replaced by the correct value 30. Then the corrected variance is

- (1) 78.00 (2) 188.66 (3) 177.33 (4) 8.33

2. In a series of $2n$ observations, half of them equal a and remaining half equal $-a$. If the standard deviation of the observations is 2, then $|a|$ equals

- (1) $\frac{1}{n}$ (2) $\sqrt{2}$ (3) 2 (4) $\frac{\sqrt{2}}{n}$

3. Suppose a population A has 100 observations 101, 102, ..., 200 and another population B has 100 observations 151, 152, ..., 250. If V_A and V_B represent the variances of the two populations, respectively, then V_A/V_B is equal to

- (1) 1 (2) 9/4 (3) 4/9 (4) 2/3

4. The mean of the numbers $a, b, 8, 5$, and 10 is 6 and the variance is 6.80. Then which one of the following gives the possible values of a and b ?

- (1) $a = 0, b = 7$ (2) $a = 5, b = 2$
(3) $a = 1, b = 6$ (4) $a = 3, b = 4$

5. Let x_1, x_2, x_3, x_4, x_5 be the observations with mean m and standard deviation s . The standard deviation of the observations $kx_1, kx_2, kx_3, kx_4, kx_5$ is

- (1) $k + s$ (2) $\frac{s}{k}$ (3) ks (4) s

6. Let $x_1, x_2, \dots, x_n$ be n observations. Let $w_i = lx_i + k$ for $i = 1, 2, \dots, n$, where l and k are constants. The mean of x_i 's is 48 and their standard deviation is 12. Also, the mean of w_i 's is 55 and standard deviation of w_i 's is 15. The values of l and k should be

- (1) $l = 2.5, k = 5$ (2) $l = -1.25, k = 5$
(3) $l = 2.5, k = -5$ (4) $l = 1.25, k = -5$

ANSWERS

1. (1) 2. (3) 3. (1) 4. (4) 5. (3)
6. (4)

Exercises

Single Correct Answer Type

1. Coefficients of variation of two distributions are 50 and 60, and their arithmetic means are 30 and 25, respectively. Difference of their standard deviations is

(1) 0 (2) 1 (3) 1.5 (4) 2.5

2. The mean of a set of numbers is $\bar{X}$. If each number is divided by 3, then the new mean is

(1) $\bar{X}$ (2) $\bar{X} + 3$ (3) $3\bar{X}$ (4) $\frac{\bar{X}}{3}$

3. The AM of the series 1, 2, 4, 8, 16, ..., 2^n is

(1) $\frac{2^n - 1}{n}$ (2) $\frac{2^{n+1} - 1}{n + 1}$

(3) $\frac{2^n + 1}{n}$ (4) $\frac{2^n - 1}{n + 1}$

4. The mean of n items is $\bar{X}$. If the first item is increased by 1, second by 2 and so on, then the new mean is

(1) $\bar{X} + n$ (2) $\bar{X} + \frac{n}{2}$
(3) $\bar{X} + \frac{n+1}{2}$ (4) None of these

5. In a moderately skewed distribution, the values of mean and median are 5 and 6, respectively. The value of mode in such a situation is approximately equal to

(1) 8 (2) 11
(3) 6 (4) None of these

6. For a normal distribution if the mean is M , mode is M_0 and median is M_d , then

(1) $M > M_d > M_0$ (2) $M < M_d < M_0$
(3) $M = M_d = M_0$ (4) $M = M_d = M_0$

7. The following data give the distribution of heights of students:

Height (in cm)	160	150	152	161	156	154	155
Number of students	12	8	4	4	3	3	7

The median of the distribution is

(1) 154 (2) 155 (3) 160 (4) 161

8. An automobile driver travels from a plain to a hill station 120 km away at an average speed of 30 km per hour. He then makes the return trip at an average speed of 25 km per hour. He covers another 120 km on the plain at an average speed of 50 km per hour. His average speed (in km/hr) over the entire distance of 360 km will be

(1) $\frac{30 + 25 + 50}{3}$ (2) $\frac{\frac{1}{30} + \frac{1}{25} + \frac{1}{50}}{3}$
(3) $\frac{3}{\frac{1}{30} + \frac{1}{25} + \frac{1}{50}}$ (4) None of these

9. The mean deviation of the data 3, 10, 10, 4, 7, 10, 5 from the mean is

(1) 2 (2) 2.57 (3) 3 (4) 3.75

10. When tested, the lives (in hours) of 5 bulbs were noted as follows:

1357, 1090, 1666, 1494, 1623

The mean deviation (in hours) from their mean is

(1) 178 (2) 179 (3) 220 (4) 356

11. Following are the marks obtained by 9 students in a mathematics test:

50, 69, 20, 33, 53, 39, 40, 65, 59

The mean deviation from the median is

(1) 9 (2) 10.5 (3) 12.67 (4) 14.76

12. If the mean of the distribution is 2.6, then the value of y is

Variate x	1	2	3	4	5
Frequency f of x	4	5	y	1	2

(1) 24 (2) 13 (3) 8 (4) 3

13. If the mean of the set of numbers $x_1, x_2, x_3, \dots, x_n$ is $\bar{x}$, then the mean of the numbers $x_i + 2i, 1 \leq i \leq n$ is

(1) $\bar{x} + 2n$ (2) $\bar{x} + n + 1$ (3) $\bar{x} + 2$ (4) $\bar{x} + n$

14. The harmonic mean of 4, 8, 16 is

(1) 6.4 (2) 6.7 (3) 6.85 (4) 7.8

15. The average of n numbers $x_1, x_2, x_3, \dots, x_n$ is M . If x_n is replaced by x' , then new average is

(1) $M - x_n + x'$ (2) $\frac{nM - x_n + x'}{n}$
(3) $\frac{(n-1)M + x'}{n}$ (4) $\frac{M - x_n + x'}{n}$

16. The following data gives the distribution of height of students:

Height (in cm)	160	150	152	161	156	154	155
Number of students	12	8	4	4	3	3	7

The median of the distribution is

(1) 154 (2) 155 (3) 160 (4) 161

17. For a slightly asymmetric distribution, mean and median are 5 and 6, respectively. What is its mode?

(1) 5 (2) 6 (3) 7 (4) 8

18. Runs scored by a batsman in 10 innings are: 38, 70, 48, 34, 42, 55, 63, 46, 54, 44
The mean deviation is
(1) 8.6 (2) 6.4 (3) 10.6 (4) 9.6
19. If μ is the mean of distribution (y_i, f_i) , then $\sum f_i (y_i - \mu) =$
(1) M.D. (2) S.D.
(3) 0 (4) Relative frequency
20. The range of the following set of observations 2, 3, 5, 9, 8, 7, 6, 5, 7, 4, 3 is
(1) 11 (2) 7 (3) 5.5 (4) 6
21. If each observation of a raw data whose variance is σ is multiplied by h , then the variance of the new set is
(1) σ^2 (2) $h^2 \sigma^2$ (3) $h \sigma^2$ (4) $h + \sigma^2$
22. If a variable x takes values 0, 1, 2, ..., n with frequencies proportional to the binomial coefficients ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$, then $\text{var}(X)$ is
(1) $\frac{n^2 - 1}{12}$ (2) $\frac{n}{2}$
(3) $\frac{n}{4}$ (4) None of these
23. The variance of the data 2, 4, 6, 8, 10 is
(1) 6 (2) 7
(3) 8 (4) None of these
24. If the standard deviation of 0, 1, 2, 3, ..., 9 is K , then the standard deviation of 10, 11, 12, 13, ..., 19 is
(1) K (2) $K + 10$
(3) $K + \sqrt{10}$ (4) $10K$
25. For a given distribution of marks, the mean is 35.16 and its standard deviation is 19.76. The coefficient of variation is
(1) $\frac{35.16}{19.76}$ (2) $\frac{19.76}{35.16}$
(3) $\frac{35.16}{19.76} \times 100$ (4) $\frac{19.76}{35.16} \times 100$
26. The mean and the SD of 1, 2, 3, 4, 5, 6 are
(1) $\frac{7}{2}, \sqrt{\frac{35}{12}}$ (2) 3, 3
(3) $\frac{7}{2}, \sqrt{3}$ (4) 3, $\frac{35}{12}$
27. The standard deviation of 25 numbers is 40. If each of the numbers is increased by 5, then the new standard deviation will be
(1) 40 (2) 45
(3) $40 + \frac{21}{25}$ (4) None of these
28. Consider any set of observations $x_1, x_2, x_3, \dots, x_{101}$. It is given that $x_1 < x_2 < x_3 < \dots < x_{100} < x_{101}$; then the mean deviation of this set of observations about a point k is minimum when k equals
(1) x_1 (2) x_{51}
(3) $\frac{x_1 + x_2 + \dots + x_{101}}{101}$ (4) x_{50}
29. For $(2n + 1)$ observations $x_1, -x_1, x_2, -x_2, \dots, x_n, -x_n$ and 0, where all x 's are distinct, let SD and MD denote the standard deviation and median, respectively. Then which of the following is always true?
(1) $SD < MD$
(2) $SD > MD$
(3) $SD = MD$
(4) Nothing can be said in general about the relationship between SD and MD
30. Let r be the range and $S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ be the SD of a set of observations $x_1, x_2, \dots, x_n$, then
(1) $S \leq r \sqrt{\frac{n}{n-1}}$ (2) $S = r \sqrt{\frac{n}{n-1}}$
(3) $S \geq r \sqrt{\frac{n}{n-1}}$ (4) None of these
31. The SD of a variate x is σ . The SD of the variate $(ax + b)/c$, where a, b, c are constants, is
(1) $\left(\frac{a}{c}\right)\sigma$ (2) $\left|\frac{a}{c}\right|\sigma$
(3) $\left(\frac{a^2}{c^2}\right)\sigma$ (4) None of these
32. The standard deviation of the data 6, 5, 9, 13, 12, 8, 10 is
(1) $\sqrt{\frac{52}{7}}$ (2) $\frac{52}{7}$ (3) $\sqrt{6}$ (4) 6
33. The mean of 100 observations is 50 and their standard deviation is 5. The sum of squares of all the observations is
(1) 50000 (2) 250000 (3) 252500 (4) 255000
34. Standard deviations for first 10 natural numbers is
(1) 5.5 (2) 3.87 (3) 2.97 (4) 2.87
35. Consider the numbers 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10. If 1 is added to each number, the variance of the numbers so obtained is
(1) 6.5 (2) 2.87 (3) 3.87 (4) 8.25
36. Consider the first 10 positive integers. If we multiply each number by -1 and then add 1 to each number, the variance of the numbers so obtained is
(1) 8.25 (2) 6.5 (3) 3.87 (4) 2.87
37. The following information relates to a sample of size 60: $\sum x^2 = 18000, \sum x = 960$. The variance is
(1) 6.63 (2) 16 (3) 22 (4) 44
38. The standard deviation of some temperature data in $^{\circ}\text{C}$ is 5. If the data were converted into $^{\circ}\text{F}$, the new variance would be
(1) 81 (2) 57 (3) 36 (4) 25
39. What is the standard deviation of the following data?
- | | | | | |
|-------------|------|-------|-------|-------|
| Measurement | 0-10 | 10-20 | 20-30 | 30-40 |
| Frequency | 1 | 3 | 4 | 2 |
- (1) 81 (2) 7.6 (3) 9 (4) 2.26

Answers Key

EXERCISES

Single Correct Answer Type

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (1) | 2. (4) | 3. (2) | 4. (3) | 5. (1) | 16. (2) | 17. (4) | 18. (1) | 19. (3) | 20. (2) |
| 6. (4) | 7. (2) | 8. (3) | 9. (2) | 10. (1) | 21. (2) | 22. (3) | 23. (3) | 24. (1) | 25. (4) |
| 11. (3) | 12. (3) | 13. (2) | 14. (3) | 15. (2) | 26. (1) | 27. (1) | 28. (2) | 29. (2) | 30. (1) |
| | | | | | 31. (2) | 32. (1) | 33. (3) | 34. (4) | 35. (4) |
| | | | | | 36. (1) | 37. (4) | 38. (1) | 39. (3) | |

Chapter 11

Concept Application Exercises

Exercise 11.1

1. (2) Total students = 100
Total marks of 70 students is $75 \times 70 = 5250$.
Total marks of girls is $7200 - 5250 = 1950$
So average of girls is
$$\frac{1950}{30} = 65$$
2. (4) Since $n = 9$, the median term is the
 $\left(\frac{9+1}{2}\right)^{\text{th}} = 5^{\text{th}}$ term.
Last four observations are increased by 2.
Since the median is the 5^{th} observation, which remains unchanged, there will be no change in the median.
3. (4) Mode + 2 Mean = 3 Median
 $\Rightarrow \text{Mode} = 3 \times 22 - 2 \times 21 = 66 - 42 = 24$
4. (3) $52x + 42y = 50(x + y)$
 $\Rightarrow 2x = 8y$
 $\Rightarrow \frac{x}{y} = \frac{4}{1}$ and $\frac{x}{x+y} = \frac{4}{5}$
The percentage of boys is 80.
5. (1)

Marks obtained	No. of students	Cumulative frequency
0-10	2	2
10-20	18	20
20-30	30	50
30-40	45	95
40-50	35	130
50-60	20	150
60-70	6	156
70-80	3	159

$N = \Sigma f = 159$ (Odd number)

Median is $\frac{1}{2}(n+1) = \frac{1}{2}(159+1) = 80^{\text{th}}$ value, which lies in the class 30-40 (see the row of cumulative frequency 95, which contains 80).

Hence median class is 30-40.

$\therefore$ We have

l = Lower limit of median class = 30

f = Frequency of median class = 45

C = Total of all frequencies preceding median class = 50

i = Width of class interval of median class = 10

$$\begin{aligned} \therefore \text{Required median} &= l + \frac{\frac{N}{2} - C}{f} \times i = 30 + \frac{\frac{159}{2} - 50}{45} \times 10 \\ &= 30 + \frac{295}{45} = 36.55 \end{aligned}$$

Exercise 11.2

1. (1) $\Sigma x = 170$, $\Sigma x^2 = 2830$
The increase in Σx is 10, then
 $\Sigma x' = 170 + 10 = 180$
The increase in Σx^2 is $900 - 400 = 500$, then
 $\Sigma x'^2 = 2830 + 500 = 3330$
Variance = $\frac{1}{n} \Sigma x'^2 - \left(\frac{1}{n} \Sigma x'\right)^2$
 $= \frac{1}{15} \times 3330 - \left(\frac{1}{15} \times 180\right)^2$
 $= 222 - 144 = 78$
2. (3) In the $2n$ observations, half of them equal to a and the remaining half equal to $-a$. Then the mean of total $2n$ observations is equal to zero. Therefore,
$$\text{S.D.} = \sqrt{\frac{\Sigma(x-x')^2}{N}}$$

$$2 = \sqrt{\frac{\Sigma x'^2}{2n}}$$

$$\Rightarrow 4 = \frac{\Sigma x'^2}{2n} \Rightarrow 4 = \frac{2a^2}{2n} \Rightarrow a^2 = 4$$

$$\Rightarrow |a| = 2$$

3. (1) $\sigma_x^2 = \frac{\Sigma d_i^2}{n}$ (Here deviations are taken from the mean.)

Since both A and B have 100 consecutive integers, they have the same standard deviation and hence variance. Therefore,

$$\frac{V_A}{V_B} = 1 \quad (\text{As } \Sigma d_i^2 \text{ is same in both the cases.})$$

4. (4) The mean of $a, b, 8, 5$, and 10 is 6 .

$$\frac{a + b + 8 + 5 + 10}{5} = 6$$

$$\Rightarrow a + b = 7 \quad (1)$$

Given that the variance is 6.8 .

$$\begin{aligned} \text{Variance} &= \frac{\Sigma(X_i - A)^2}{n} \\ &= \frac{(a-6)^2 + (b-6)^2 + 4 + 1 + 16}{5} = 6.8 \end{aligned}$$

$$\begin{aligned} \Rightarrow a^2 + b^2 &= 25 \\ \Rightarrow a^2 + (7-a)^2 &= 25 & [\text{Using (1)}] \\ \Rightarrow a^2 - 7a + 12 &= 0 \\ \therefore a &= 4, 3 \text{ and } b = 3, 4 \end{aligned}$$

5. (3) Here, $m = \frac{\Sigma x_i}{5}$, $s = \sqrt{\frac{\Sigma x_i^2}{5} - \left(\frac{\Sigma x_i}{5}\right)^2}$

So, for observations $kx_1, kx_2, kx_3, kx_4, kx_5$,

$$\begin{aligned} \text{S.D.} &= \sqrt{\frac{k^2 \Sigma x_i^2}{5} - \left(\frac{k \Sigma x_i}{5}\right)^2} \\ &= \sqrt{\frac{k^2 \Sigma x_i^2}{5} - k^2 \left(\frac{\Sigma x_i}{5}\right)^2} \\ &= k \sqrt{\left(\frac{\Sigma x_i^2}{5}\right) - \left(\frac{\Sigma x_i}{5}\right)^2} = ks \end{aligned}$$

6. (4) Given, $w_i = lx_i + k$, $\bar{x}_i = 48$, S.D. of x_i 's = 12, $w_i = 55$ and S.D. of w_i 's = 15

Then, $\bar{w}_i = l\bar{x}_i + k$ [where, $\bar{w}_i$ is mean of w_i 's
and $\bar{x}_i$ is mean of x_i 's]

$$\Rightarrow 55 = 48l + k \quad \dots (i)$$

Now, S.D. of w_i 's = $l \times$ (S.D. of x_i 's)

$$\Rightarrow 15 = l \times 12$$

$$\Rightarrow l = \frac{15}{12} = 1.25$$

From Eq. (i), we get

$$k = 55 - 1.25 \times 48 = -5$$

Exercises

Single Correct Answer Type

1. (1) Here, $(CV)_1 = 50$, $(CV)_2 = 60$, $\bar{x}_1 = 30$ and $\bar{x}_2 = 25$

$$\therefore (CV)_1 = \frac{\sigma_1}{\bar{x}_1} \times 100$$

$$\Rightarrow 50 = \frac{\sigma_1}{30} \times 100$$

$$\therefore \sigma_1 = \frac{30 \times 50}{100} = 15$$

$$\text{and } (CV)_2 = \frac{\sigma_2}{\bar{x}_2} \times 100$$

$$\Rightarrow 60 = \frac{\sigma_2}{25} \times 100$$

$$\therefore \sigma_2 = \frac{60 \times 25}{100} = 15$$

$$\text{Now, } \sigma_1 - \sigma_2 = 15 - 15 = 0$$

2. (4) Let $x_1, x_2, \dots, x_n$ be n numbers. Then $\bar{X} = \frac{1}{n} \sum_{i=1}^n x_i$.

If each number is divided by 3, then the new mean $\bar{Y}$ is given by

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n \left(\frac{x_i}{3} \right) = \frac{1}{3} \left(\frac{1}{n} \sum_{i=1}^n x_i \right) = \frac{\bar{X}}{3}$$

3. (2) The required AM is

$$\bar{X} = \frac{1 + 2 + 2^2 + 2^3 + \dots + 2^n}{n+1} = \frac{(2^{n+1} - 1)}{(n+1)(2-1)} = \frac{2^{n+1} - 1}{n+1}$$

4. (3) Let $x_1, x_2, \dots, x_n$ be n items. Then $\bar{X} = \frac{1}{n} \sum x_i$. Let $y_1 = x_1 + 1, y_2 = x_2 + 2, y_3 = x_3 + 3, \dots, y_n = x_n + n$.

Then the mean of the new series is

$$\begin{aligned} \frac{1}{n} \sum (x_i + i) &= \frac{1}{n} \sum x_i + \frac{1}{n} (1 + 2 + 3 + \dots + n) \\ &= \bar{X} + \frac{1}{n} \frac{n(n+1)}{2} \\ &= \bar{X} + \frac{n+1}{2} \end{aligned}$$

5. (1) For a moderately skewed distribution, we have

$$\text{Mode} = 3 \text{ median} - 2 \text{ mean}$$

$$\Rightarrow \text{Mode} = 3(6) - 2(5) = 8.$$

6. (4) It is obvious.

7. (2) Arranging the data in ascending order of magnitude, we obtain

Height (in cm)	150	152	154	155	156	160	161
Number of students	8	4	3	7	3	12	4
Cumulative frequency	8	12	15	22	25	37	41

Here, the total number of items is 41, i.e., an odd number.

Hence, the median is $\frac{41+1}{2}$ th i.e., 21st item.

From cumulative frequency table, we find that the median, i.e. 21st item is 155.

(All items from 16th to 22nd are equal, each 155).

8. (3) Average speed = $\frac{120 + 120 + 120}{\frac{120}{30} + \frac{120}{25} + \frac{120}{50}}$

$$= \frac{3}{\frac{1}{30} + \frac{1}{25} + \frac{1}{50}} \text{ km/hour}$$

9. (2) Given, observations are 3, 10, 10, 4, 7, 10 and 5.

$$\therefore \bar{x} = \frac{3 + 10 + 10 + 4 + 7 + 10 + 5}{7} = \frac{49}{7} = 7$$

x_i	$d_i = x_i - \bar{x} $
3	4
10	3
10	3
4	3
7	0
10	3
5	2
Total	$\Sigma d_i = 18$

$$\text{Now, M.D.} = \frac{\Sigma d_i}{N} = \frac{18}{7} = 2.57$$

10. (1) The lives of 5 bulbs are: 1357, 1090, 1666, 1494, 1623

$$\begin{aligned} \text{Mean } (\bar{x}) &= \frac{1357 + 1090 + 1666 + 1494 + 1623}{5} \\ &= \frac{7230}{5} = 1446 \end{aligned}$$

x_i	$d_i = x_i - \bar{x} $
1357	89
1090	356
1666	220
1494	48
1623	177
Total	$\Sigma d_i = 890$

$$\text{M.D.} = \frac{\Sigma d_i}{N} = \frac{890}{5} = 178$$

11. (3) Marks obtained by 9 students in Mathematics are: 50, 69, 20, 33, 53, 39, 40, 65, 59

Let us rewrite the given data in ascending order.

20, 33, 39, 40, 50, 53, 59, 65, 69,

Here, $n = 9$

[odd]

Median (M_e) = 5th term = 50

x_i	$d_i = x_i - M_e $
20	30
33	17
39	11
40	10
50	0
53	3
59	9
65	15
69	19
N = 2	$\Sigma d_i = 114$

$$\text{M.D.} = \frac{114}{9} = 12.67$$

$$12. (3) \text{ Mean} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i}$$

$$\therefore 2.6 = \frac{1 \times 4 + 2 \times 5 + 3 \times y + 4 \times 1 + 5 \times 2}{4 + 5 + y + 1 + 2}$$

$$\Rightarrow 31.2 + 2.6y = 28 + 3y$$

$$\Rightarrow 0.4y = 3.2$$

$$\Rightarrow y = 8$$

$$13. (2) \text{ We know that } \bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$\Rightarrow \sum_{i=1}^n x_i = n\bar{x}$$

$$\therefore \frac{\sum_{i=1}^n (x_i + 2i)}{n} = \frac{\sum_{i=1}^n x_i + 2 \sum_{i=1}^n i}{n} = \frac{n\bar{x} + 2(1 + 2 + \dots + n)}{n}$$

$$= \frac{n\bar{x} + 2 \frac{n(n+1)}{2}}{n} = \bar{x} + (n+1)$$

$$14. (3) \text{ H.M. of } 4, 8, 16 = \frac{3}{\frac{1}{4} + \frac{1}{8} + \frac{1}{16}} = \frac{48}{7} = 6.85$$

$$15. (2) M = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

$$\Rightarrow nM = x_1 + x_2 + x_3 + \dots + x_{n-1} + x_n$$

$$\Rightarrow nM - x_n = x_1 + x_2 + x_3 + \dots + x_{n-1}$$

$$\Rightarrow nM - x_n + x' = x_1 + x_2 + x_3 + \dots + x_{n-1} + x'$$

$$\Rightarrow \frac{nM - x_n + x'}{n} = \frac{x_1 + x_2 + x_3 + \dots + x_{n-1} + x'}{n}$$

$$\therefore \text{New average} = \frac{nM - x_n + x'}{n}$$

16. (2) Arranging the data in ascending order of magnitude, we obtain

Height (in cm)	150	152	154	155	156	160	161
Number of students	8	4	3	7	3	12	4
Cumulative frequency	8	12	15	22	25	37	41

Here, total number of items is 41, which is an odd number.

Hence, the median is $\frac{41+1}{2}$ th = 21st item.

From cumulative frequency table, we find that median i.e., 21st item is 155.

17. (4) We know that

$$\text{Mode} = (3 \times \text{Median}) - (2 \times \text{Mean}) = 3(6) - 2(5) = 8$$

18. (1) Arranging the given data in ascending order, we have

34, 38, 42, 44, 46, 48, 54, 55, 63, 70,

$$\text{Median, } M = \frac{46 + 48}{2} = 47 \quad (\because n = 10, \text{ median is the mean of 5th and 6th items})$$

$$\therefore \text{Mean deviation} = \frac{\sum |x_i - M|}{n}$$

$$= \frac{\sum |x_i - 47|}{10}$$

$$= \frac{13 + 9 + 5 + 3 + 1 + 1 + 7 + 8 + 16 + 23}{10}$$

$$= 8.6$$

19. (3) We have,

$$\sum f_i (y_i - \mu) = \sum f_i y_i - \mu \sum f_i = \mu \sum f_i - \mu \sum f_i = 0$$

$$\left[\because \mu = \frac{\sum f_i y_i}{\sum f_i} \right]$$

$$20. (2) \text{ Range} = X_{\text{highest}} - X_{\text{lowest}} = 9 - 2 = 7$$

21. (2) Let $x_1, x_2, \dots, x_n$ be the raw data. Then,

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{X})^2$$

If each value is multiplied by h , then the values become $hx_1,$

$hx_2, \dots, hx_n$. Then AM of the values is

$$\frac{hx_1 + hx_2 + \dots + hx_n}{n} = h\bar{X}$$

Therefore, the variance of the new set of values is

$$\frac{1}{n} \sum_{i=1}^n (hx_i - h\bar{X})^2 = h^2 \left(\frac{1}{n} \sum_{i=1}^n (x_i - \bar{X})^2 \right) = h^2 \sigma^2$$

22. (3) We have

$$\bar{X} = \frac{0^n C_0 + 1^n C_1 + 2^n C_2 + \dots + n^n C_n}{n C_0 + n C_1 + n C_2 + \dots + n C_n} = \frac{\sum_{r=0}^n r^n C_r}{\sum_{r=0}^n n C_r}$$

$$= \frac{1}{2^n} \sum_{r=1}^n r \frac{n}{r} {}^{n-1}C_{r-1}$$

$$\left[\because \sum_{r=0}^n {}^n C_r = 2^n; {}^n C_r = \frac{n}{r} {}^{n-1}C_{r-1} \right]$$

$$= \frac{n}{2^n} \sum_{r=1}^n {}^{n-1}C_{r-1} = \frac{n}{2^n} 2^{n-1} = \frac{n}{2} \left[\because \sum_{r=1}^n {}^{n-1}C_{r-1} = 2^{n-1} \right]$$

$$\begin{aligned} \text{and } \frac{1}{N} \sum f_i x_i^2 &= \frac{1}{2^n} \sum r^2 {}^nC_r = \frac{1}{2^n} \sum_{r=0}^n [r(r-1) + r] {}^nC_r \\ &= \frac{1}{2^n} \left\{ \sum_{r=0}^n r(r-1) {}^nC_r + \sum_{r=0}^n r {}^nC_r \right\} \\ &= \frac{1}{2^n} \left\{ \sum_{r=2}^n r(r-1) \frac{n}{r} \frac{n-1}{r-1} {}^{n-2}C_{r-2} + \sum_{r=1}^n r \frac{n}{r} {}^{n-1}C_{r-1} \right\} \\ &= \frac{1}{2^n} \{n(n-1) 2^{n-2} + n 2^{n-1}\} = \frac{n(n-1)}{4} + \frac{n}{2} \end{aligned}$$

$$\therefore \text{Var}(X) = \frac{1}{N} \sum f_i x_i^2 - \bar{X}^2 = \frac{n(n-1)}{4} + \frac{n}{2} - \frac{n^2}{4} = \frac{n}{4}$$

23. (3) Here, $\bar{x} = \frac{2+4+6+8+10}{5} = 6$

$$\begin{aligned} \text{Hence, variance} &= \frac{1}{n} \sum (x_i - \bar{x})^2 \\ &= \frac{1}{5} \{(2-6)^2 + (4-6)^2 + (6-6)^2 + (8-6)^2 + (10-6)^2\} \\ &= \frac{1}{5} \{16 + 4 + 0 + 4 + 16\} = \frac{1}{5} \{40\} = 8 \end{aligned}$$

24. (1) It is obvious.

25. (4) Coefficient of variation = $\frac{\text{SD}}{\text{Mean}} \times 100 = \frac{19.76}{35.16} \times 100$

26. (1) Mean $\bar{x} = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = \frac{7}{2}$

$$\begin{aligned} \text{SD} = \sigma &= \sqrt{\frac{1}{n} \sum_i^2 - (\bar{x})^2} \\ &= \sqrt{\frac{1}{6}(1+4+9+16+25+36) - \frac{49}{4}} \\ &= \sqrt{\frac{91}{6} - \frac{49}{4}} = \sqrt{\frac{182-147}{12}} = \sqrt{\frac{35}{12}} \end{aligned}$$

27. (1) If each item of the data is increased or decreased by the same constant, the standard deviation of the data remains unchanged.

28. (2) Mean deviation is minimum when it is considered about an item equidistant from the beginning and the end, i.e., the median. In this case, the median is $\frac{101+1}{2}$ th, i.e., 51st item, i.e., x_{51} .

29. (2) On arranging the given observations in ascending order, we get

All negative terms $\underbrace{0}_{(n+1)\text{th term}}$ All positive terms

Median of given observations = $(n+1)$ th term = 0

$\therefore \text{SD} > \text{MD}$

30. (1) We have $r = \max |x_i - x_j|$

$$\text{and } S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$\text{Now, } (x_i - \bar{x})^2 = \left(x_i - \frac{x_1 + x_2 + \dots + x_n}{n} \right)^2$$

$$\begin{aligned} &= \frac{1}{n^2} [(x_i - x_1) + (x_i - x_2) + \dots + (x_i - x_i - 1) \\ &\quad + (x_i - x_1 + 1) + \dots + (x_i - x_n)] \leq \frac{1}{n^2} [(n-1)r]^2, \\ &\quad [\because |x_i - x_j| \leq r] \end{aligned}$$

$$\Rightarrow (x_i - \bar{x})^2 \leq r^2$$

$$\Rightarrow \sum_{i=1}^n (x_i - \bar{x})^2 \leq nr^2$$

$$\Rightarrow \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \leq \frac{nr^2}{(n-1)}$$

$$\Rightarrow S^2 \leq \frac{nr^2}{(n-1)}$$

$$\Rightarrow S \leq r \sqrt{\frac{n}{n-1}}$$

31. (2) Let $y = \frac{ax+b}{c}$, i.e., $y = \frac{a}{c}x + \frac{b}{c}$

$$\text{i.e., } y = \frac{a}{c}x + \frac{b}{c}$$

$$\therefore \bar{y} = A\bar{x} + B$$

$$\therefore y - \bar{y} = A(x - \bar{x})$$

$$\Rightarrow (y - \bar{y})^2 = A^2 (x - \bar{x})^2$$

$$\Rightarrow \Sigma(y - \bar{y})^2 = A^2 \Sigma(x - \bar{x})^2$$

$$\Rightarrow n \cdot \sigma_y^2 = A^2 \cdot n \sigma_x^2$$

$$\Rightarrow \sigma_y^2 = A^2 \sigma_x^2$$

$$\Rightarrow \sigma_y = |A| \sigma_x$$

$$\Rightarrow \sigma_y = \left| \frac{a}{c} \right| \sigma_x$$

$$\text{Thus, new SD} = \left| \frac{a}{c} \right| \sigma$$

32. (1) Given, numbers are: 6, 5, 9, 13, 12, 8, 10

x_i	x_i^2
6	36
5	25
9	81
13	169
12	144
8	64
10	100
$\Sigma x_i = 63$	$\Sigma x_i^2 = 619$

$$\begin{aligned} \therefore \text{S.D.} = \sigma &= \sqrt{\frac{\Sigma x_i^2}{N} - \left(\frac{\Sigma x_i}{N} \right)^2} = \sqrt{\frac{619}{7} - \left(\frac{63}{7} \right)^2} \\ &= \sqrt{\frac{7 \times 619 - 3969}{49}} = \sqrt{\frac{4333 - 3969}{49}} \\ &= \sqrt{\frac{364}{49}} = \sqrt{\frac{52}{7}} \end{aligned}$$

33. (3) Given, $\bar{x} = 50$, $n = 100$ and $\sigma = 5$

$$\Rightarrow 50 = \frac{\Sigma x_i}{100}$$

$$\Rightarrow \Sigma x_i = 50 \times 100 = 5000$$

$$\text{Now, } s^2 = \frac{\Sigma x_i^2}{n} - (\bar{x})^2$$

$$\Rightarrow 25 = \frac{\Sigma x_i^2}{100} - (50)^2$$

$$\Rightarrow 2525 = \frac{\Sigma x_i^2}{100}$$

$$\therefore \Sigma x_i^2 = 252500$$

34. (4) We know that, SD of first n natural numbers = $\sqrt{\frac{n^2 - 1}{12}}$

$$\begin{aligned} \therefore \text{SD of first 10 numbers} &= \sqrt{\frac{10^2 - 1}{12}} \\ &= \sqrt{\frac{100 - 1}{12}} = \sqrt{\frac{99}{12}} = \sqrt{8.25} = 2.87 \end{aligned}$$

35. (4) Given numbers are 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10.

If 1 is added to each number, then observations will be 2, 3, 4, 5, 6, 7, 8, 9, 10 and 11.

$$\begin{aligned} \therefore \Sigma x_i &= 2 + 3 + 4 + \dots + 11 \\ &= \frac{10}{2} [2 \times 2 + 9 \times 1] = 5[4 + 9] = 65 \end{aligned}$$

$$\begin{aligned} \text{and } \Sigma x_i^2 &= 2^2 + 3^2 + 4^2 + 5^2 + \dots + 11^2 \\ &= (1^2 + 2^2 + 3^2 + \dots + 11^2) - (1^2) \\ &= \frac{11 \times 12 \times 23}{6} - 1 = 505 \end{aligned}$$

$$\begin{aligned} \therefore s^2 &= \frac{\Sigma x_i^2}{n} - \left(\frac{\Sigma x_i}{n} \right)^2 = \frac{505}{10} - \left(\frac{65}{10} \right)^2 \\ &= 50.5 - (6.5)^2 \\ &= 50.5 - 42.25 \\ &= 8.25 \end{aligned}$$

36. (1) The first 10 positive integers are 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10.

On multiplying each number by -1 , we get

$$-1, -2, -3, -4, -5, -6, -7, -8, -9, -10$$

On adding 1 to each number, we get

$$0, -1, -2, -3, -4, -5, -6, -7, -8, -9$$

$$\therefore \Sigma x_i = -\frac{9 \times 10}{2} = -45$$

$$\text{and } \Sigma x_i^2 = 0^2 + (-1)^2 + (-2)^2 + \dots + (-9)^2 = \frac{9 \times 10 \times 19}{6} = 285$$

$$\begin{aligned} \therefore \text{S.D.} &= \sqrt{\frac{285}{10} - \left(\frac{-45}{10} \right)^2} = \sqrt{\frac{285}{10} - \frac{2025}{100}} \\ &= \sqrt{\frac{2850 - 2025}{100}} = \sqrt{8.25} \end{aligned}$$

$$\text{Now, variance} = (\text{S.D.})^2 = (\sqrt{8.25})^2 = 8.25$$

$$\begin{aligned} 37. (4) \text{ Variance} &= \frac{\Sigma x_i^2}{n} - \left(\frac{\Sigma x_i}{n} \right)^2 \\ &= \frac{18000}{60} - \left(\frac{960}{60} \right)^2 = 300 - 256 = 44 \end{aligned}$$

38. (1) Given, $\sigma_C = 5$

Relation between $^{\circ}\text{C}$ and $^{\circ}\text{F}$ is given by

$$F = \frac{9C}{5} + 32$$

$$\therefore \sigma_F = \frac{9}{5} \sigma_C = \frac{9}{5} \times 5 = 9$$

$$\text{Here, } \sigma_F^2 = (9)^2 = 81$$

39. (3)

Class	Frequency	y_i	$u_i = \frac{y_i - A}{10},$ $A = 25$	$f_i u_i$	$f_i u_i^2$
0-10	1	5	-2	-2	4
10-20	3	15	-1	-3	3
20-30	4	25	0	0	0
30-40	2	35	1	2	2
Total	10			-3	9

$$\sigma^2 = c^2 \left[\frac{\Sigma f_i u_i^2}{\Sigma f_i} - \left(\frac{\Sigma f_i u_i}{\Sigma f_i} \right)^2 \right]$$

$$= 10^2 \left[\frac{9}{10} - \left(\frac{-3}{10} \right)^2 \right] = 90 - 9 = 81$$

$$\Rightarrow \sigma = 9$$